

Load Balancing Method Using Server Temperature Prediction Considering Multiple Internal Heat Sources in Data Centers

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Abstract—The increasing demand for cloud computing services to more flexibly use computing resources has led to a severe problem of excessive energy consumption in data centers. In recent research, the energy consumption of computing and cooling equipment has attracted much attention. To reduce the energy consumption in data centers, the energy efficiency of the cooling equipment needs to be considered. This study theoretically modeled the relations between the server's internal heat sources and exhaust temperatures. First, we proposed a model that expands the previous study to describe the temperature change delay and to apply it to multiple internal heat sources. Then, we experimentally evaluated the prediction accuracy of the proposed model using a real server and conducted a simulation to confirm the efficiency of this prediction model for load balancing. The simulation results showed that the proposed method reduced the average maximum server temperature by 0.27 °C compared to the conventional method, which theoretically leads to energy savings of 260 kW per day and 95 MW per year in typical data centers.

Keywords—data center, load balancing, server, thermal management

I. INTRODUCTION

In recent years, with the rapid emergence of internet of things (IoT), cloud computing, and other network-oriented innovations, network data traffic has rapidly increased due to various factors such as the digitization of information, popularization of smartphone tablets, and improvements in network processing ability. Figure 1 shows the transition and prediction of the world's data traffic according to Cisco. As shown in the figure, the average annual growth rate of IP traffic from 2017 to 2022 was estimated to be 122 exabytes (EB); however, it reached 396 EB by the end of 2022, which is more than three times the estimate in five years [1].

Due to this growth, the increasing amount of energy consumed by data centers has become a global problem, and data center managers are beginning to pay attention to energy consumption reduction. Out of the total energy consumption in a data center, cooling energy accounts for 30% to 50% [2]. Generally, in data centers, a computer room air conditioner (CRAC) is deployed to prevent server overheating. High temperatures usually lead to higher degradation and component failure rates and shorten the service life of IT equipment. To reduce the energy consumption and replacement costs of IT equipment in data centers, the energy efficiency of the cooling equipment should be considered.

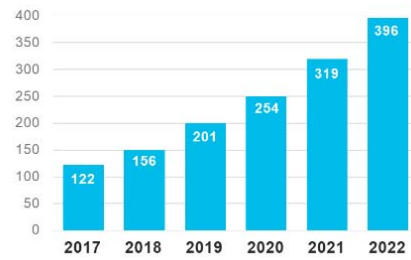


Fig.1. Global IP Traffic Growth [1]

Several studies have been conducted to investigate this problem using temperature-based load balancing to control the server temperature increase. For example, efficient temperature prediction methods for load balancing, task allocation methods based on server temperature, and server temperature predictions to improve the cooling infrastructure efficiency in data centers have been suggested. Recently, however, the computing environments in data centers have rapidly advanced and diversified. Accordingly, data centers currently provide various hardware resources and application platforms. With the spread of dedicated processors such as GPUs and the enhancement of the heterogeneous computing environment, the combined use of heterogeneous hardware should be considered in predicting temperatures in data centers. In this situation, some existing studies have focused on temperature predictions that consider the multiple resources inside servers as heat sources. However, those studies did not propose a temperature prediction method for these distributed tasks that includes the general modeling of multiple heat sources inside servers.

This study theoretically modeled the relations between the server's internal heat sources and exhaust temperatures. Moreover, the efficiency of the proposed model for load balancing was evaluated. The proposed model expands previous studies to describe the temperature change delay and multiple internal heat sources. In constructing the model, the temperature measurement system was implemented on a real server and the parameters were determined. This model enables temperature predictions for servers with different internal resources and requests with different workload characteristics. The prediction accuracy of the proposed model was evaluated experimentally using a real server, and a simulation was conducted to confirm its efficiency for load balancing. As a result, the proposed model predicted with high accuracy the server exhaust temperature for tasks with a heterogeneous workload without revising the modeling parameters. In addition, the proposed method was evaluated

through a simulation comparing it with the conventional least CPU load method, and a reduction in the server temperature variance was confirmed.

II. RELATED WORKS

In previous research, various approaches have been taken to reduce the air conditioning power consumption in data centers. In this study, we focus on the temperature management of the data center using load balancing technology and temperature prediction.

A. Load balancing technology in the data center

In the load balancing method, various algorithms corresponding to the server operating environment have been used. However, the use environment and purpose of load balancing technologies have recently diversified. One of these new uses is the application to the server temperature homogenization in the data center.

To improve the cooling efficiency, eliminating the hot spots in the data center has become a topic of interest. A data center hot spot can be described as a local temperature rise caused by the load deviation and cooling performance difference between servers. A diverse temperature distribution causes an excess operation of the cooling equipment to keep the temperature below the specified degree, and this situation increases the power consumption of the cooling systems. To moderate the power consumption, floor tiles with fans called "active fan tiles" [3] [4] and movable vent tiles [5] have been put forward as local cooling methods that are installed in front of each rack and operate individually. However, the entire CRAC system is general, and it is expensive to import cooling devices to each rack and server.

The ambient temperature (exhaust and inlet temperatures) around the server is often the focus of studies on load balancing considering the power consumption of the entire data center. Tang et al. [6] [7] proposed a task dispersion method aimed at reducing the maximum temperature of the inlet port of the server. To reduce the temperature of the inlet port and improve the cooling efficiency of the CRAC, it is necessary to control the air recirculation from the hot aisle to the cold aisle. In addition, studies have been conducted to reduce the exhaust temperature of the server by considering the CPU temperature for task allocation [8]. In this study, the strategy to assign CPU-intensive tasks to servers with lower CPU temperatures reduces the CPU maximum temperature, server exhaust temperature, and fan speed. The proposed method shows that suppressing the rise of CPU temperatures has a significant impact on the exhaust temperature decrease and cooling power of the server.

B. Temperature prediction of data center

As described in the previous section, load balancing techniques that consider temperatures are being vigorously studied. In addition, some studies are based on the predicted temperature and load balancing.

Computational fluid dynamics (CFD) has been widely used in data center temperature control. Owing to a large amount of calculations, deriving the analytical solution of the fluid equation is usually challenging. Thus, the CFD model usually uses a computer to find the solution of fluid equations numerically. In past research, CFD were used to reduce the power consumption of data center air conditioning. Arghode et al. [9] [10] used CFD to model the cooling airflow from

porous bricks in the rack unit. In addition, Athavale et al. [4] created a CFD model based on an experimental data center, and it was confirmed that the difference between the measured data and the calculated results was reduced to 1.7 °C relative to the inlet air temperature.

The computational cost of CFD is too high for the simulation of a complex fluid. Hence, some studies built simpler models to simulate the temperature increase of IT equipment in data centers, such as air conditioning setting changes, server power operation status changes, and abnormality detection. Van Gilder, Erden, and Pardey [11] [12] proposed a simple model that considers the thermal characteristics of servers. In addition, the model is extended to the rack unit, and a temperature distribution prediction simulation is performed throughout the data center. This mathematical model can be used without additional computational overhead for sudden and temporary thermal changes.

In recent years, machine learning methods have been widely used in server temperature prediction, similar to a mathematical model. In Chen's study [13], data were collected in real-time from sensors installed on the server, and an online linear regression was used to predict the server exhaust temperature [14]. There are five types of data for the input layer: sensor temperature, server fan, CRAC wind speed, CRAC set temperature, and CPU utilization. According to the predicted data, the set temperature of the CRAC, wind speed, and wind speed of the server fans are controlled to improve the temperature management efficiency. The CFD simulation results show that the cooling power consumption is reduced by 30% compared with the existing methods. In Lorenzi et al. [15], neural networks predict the temperature of the server inlet air in the rack and the temperature of the air circulating back to the CRAC. There are seven types of data input, such as server power consumption and intake mass flow.

III. PROPOSED METHOD

As mentioned above, temperature predictions using CFD simulations have been widely used in data center temperature management. However, owing to the high computational cost, this is not suitable for dynamic predictions. Therefore, a low-cost and straightforward server temperature model was proposed in [11]. In this study, the server exhaust temperature is modeled by the following equation, and Figure 2 [11] shows a diagram of the model.

$$q_{IT} = Mc_{p_{eff}} \frac{dT_{eff}}{dt} + \dot{m}c_{p_{air}}(T_{ex} - T_{amb}) \quad (1)$$

$$Mc_{p_{eff}} \frac{dT_{eff}}{dt} = hA_1(T_{amb} - T_{eff}) + hA_2(T_{amb} + \Delta T_{IT} - T_{eff}) \quad (2)$$

$$\Delta T_{IT} = \frac{q_{IT}}{\dot{m}c_{p_{air}}} \quad (3)$$

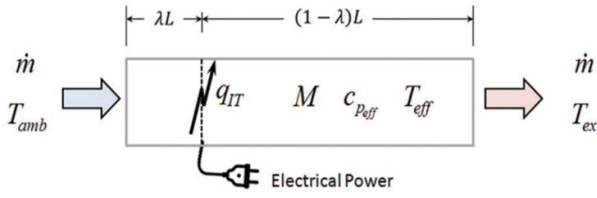


Fig.2. Server model diagram in the existing method [11]

Equation (1) describes the thermal balance between the entire server and the external environment. q_{IT} is the power consumption of the server, M is the quality of the server, $c_{p_{eff}}$ is the specific heat of the server, T_{eff} is the temperature of the server, $\dot{m}$ is the mass flow of cooling air, $c_{p_{air}}$ is the specific heat of air, and T_{ex} and T_{amb} are the exhaust and inlet temperatures of the server, respectively. Equation (2) is the energy storage formula for the fluid passing between servers and the heat exchanged between servers. h is the heat transfer coefficient, which depends on the physical characteristics and state of the flow. A represents the total surface area inside the server. In equation (3), ΔT_{IT} is the temperature rise when the incoming air passes through the heat source.

Unlike CFD, this method can predict temporary temperature changes in the server caused by sudden changes in the external air or power consumption, without an additional computational cost. However, there are some shortcomings when this model is applied to temperature predictions in load balancing. One limitation is that when the server power consumption changes suddenly owing to the load distribution, the prediction accuracy of the server temperature change is low. Another drawback is that this method cannot describe a server with multiple heat sources. It can be seen from equation (2) that in the existing method, when the power consumption of the server increases, the air temperature increase is caused only by a single heat source existing in the server. Thus, it is impossible to describe a server model with multiple heat sources.

Based on the above method, the server temperature model has the following requirements for the temperature prediction of load balancing.

- It can describe the discharging temperature delay when the load of the server changes suddenly.
- It can be extended to servers with multiple heat sources.

The following is a temperature prediction model that meets these conditions. Figure 3 shows the modeling diagram when there are two heat sources in the server.

$$q_{eff} + \sum_{k=1}^n \Delta q_k = M c_{p_{eff}} \frac{dT_{eff}}{dt} + \dot{m} c_{p_{air}} (T_{ex} - T_{amb}) + m_k c_{p_k} \sum_{k=1}^n \frac{dT_k}{dt} \quad (4)$$

$$M c_{p_{eff}} \frac{dT_{eff}}{dt} = h \left(1 - \sum_{k=1}^n \lambda_k \right) A (T_{amb} - T_{eff}) + \sum_{k=1}^n \lambda_k h A (T_{amb} + \Delta T_{lk} - T_{eff}) + q_{eff} \quad (5)$$

$$\Delta T_{lk} = \frac{h_k A_k}{\dot{m} c_{p_{air}}} (T_k - \sum_{j=1}^k \Delta T_{lj} - T_{amb}) \quad (6)$$

$$\Delta q_k = m_k c_{p_k} \frac{dT_k}{dt} + h_k A_k (T_k - T_k(0)) \quad (7)$$

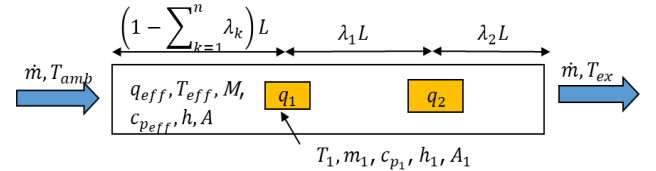


Fig.3. Server model diagram of the proposed method

Equation (4) describes the thermal balance between the entire server and the external environment. q_{eff} is the power consumption of the server chassis, Δq_k is the power consumption difference between the hardware used as a heat source and without load, n is the number of heat sources, and T_k , m_k , and c_{p_k} are the temperature, quality, and specific heat of the hardware, respectively. Equation (5) is an energy formula that represents the relationship between the fluid passing through the server and the heat exchanged with the server. h_k , A_k , and λ_k are the heat transfer coefficients of the hardware, the surface area, and their locations on the server, respectively. In equation (6), ΔT_{lk} represents the temperature increase in the incoming air as it passes through the k th heat source. Equation (7) represents the relationship between the hardware power consumption and the temperature increase.

In the previous model, all the server power consumption was converted to air heat without delay, as described in equation (3). In this model, as shown in the third item on the right side of equation (4), the server power consumption will not only lead to the air temperature increase but also to the hardware temperature increase as a heat source. Therefore, when the power consumption of the server changes, the change in the exhaust temperature will be delayed.

Similarly, in this model, multiple parts of the hardware serve as heat sources. In equation (5), the air flowing into the server is modeled as it successively passes through the heat source from the front of the server. Hence, to more realistically describe the temperature distribution inside the server, the heat generated by the power consumption of the server is determined by the q_{eff} consumption, which is evenly distributed between the server chassis and the hardware in different locations by dividing it into q_k . To simplify the model, it is assumed that the temperature of the heat source hardware varies independently of the surrounding server shell.

Next, a load balancing method using the predicted server discharge temperature is described. The purpose of load balancing using temperature predictions is to ensure the uniform temperature discharge of the server, inhibit the

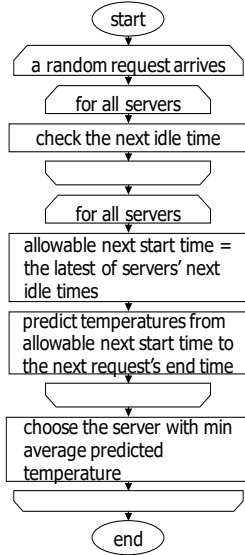


Fig.4. Flow chart of the load balancing algorithm

generation of hot spots, and improve the cooling efficiency of the CRAC. Figure 4 shows the algorithm flowchart for the proposed method. In this method, starting from the next executable time for the slowest executable-request-state server to the predetermined end time of the next request, the temperature evolution of each second is predicted. Then, the server with the lowest predicted average temperature is selected.

IV. EXPERIMENTAL ENVIRONMENT

In this study, we used a server to build the model in the laboratory server room. Table 1 lists the server specifications. Figure 5 shows the experimental environment. The red box in the figure represents the experimental server. Figure 6 shows the devices in the server. As shown in the figure, two CPUs are installed in the center of the server, six local fans are installed on the front of the server, and a GPU is installed near the back of the server.

For the evaluation of the model construction and proposed methods, the environment for obtaining the server load and temperature data is constructed. First, for the CPU and GPU loads, the utilization rates and respective power consumptions are measured. Dstat [16] is used for the CPU utilization command, and the CPU energy meter [17] is used for the CPU power consumption measurement. The GPU's utilization and power consumption are measured with the NVIDIA System Management Interface (Nvidia-smi) [18]. The integrated Lights-Out 4 (iLO4) [19] with the HP's server remote management tool are used to obtain the power consumption of the entire server every 3 s.

TABLE 1 SPECIFICATIONS OF THE SERVER AND THE EQUIPMENT INSIDE THE SERVER

	Number	Model
Server	1	HP ProLiant DL380p Gen8
Fan	6	HP 662520-001
CPU	2	Intel(R) Xeon(R) CPU E5-2665 0 @ 2.40GHz, 8 cores
GPU	1	NVIDIA Quadro P2000



Fig.5. Location of the experimental server in the rack

In addition, to build and evaluate the model, the temperature data of the CPU, GPU, memory, power supply unit (PSU), and HDD need to be obtained. For the CPU temperature, the lmsensors [20] command is used, and NVIDIA SMI is used for the GPU temperature. The temperatures of the memory and PSU are obtained by using the on-board sensor of the server through iLO4.

As for the HDD, because the sensor on the motherboard cannot measure the temperature, a K-type thermocouple sensor [21] contacts the back of the HDD body using thermal conductivity double-sided adhesive tape for bonding.

In addition to the temperature of the HDD, wireless sensor networks (WSNs) are also used to import the obtained information on the suction and exhaust temperatures of the server. The WSN is composed of Arduino [22], XBee [23], mcp9808 [24] produced by a microchip technology company,

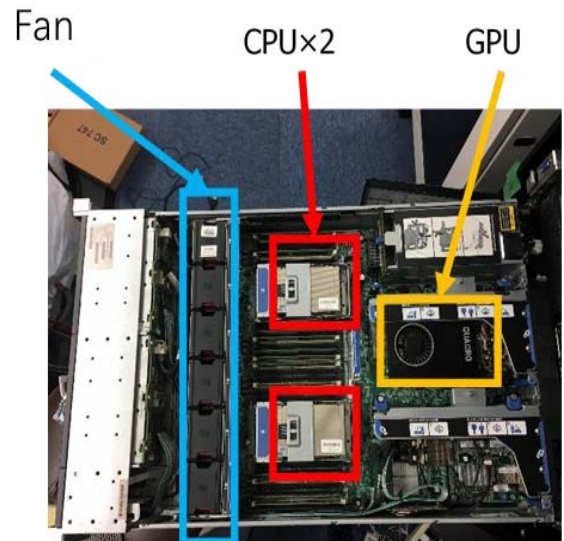


Fig.6. Device layout in the server

and a thermocouple. The data obtained by the WSN is saved in a MySQL server built on Raspberry Pi.

TABLE 2 CPU/GPU ESTIMATION PARAMETERS

	$m_k c_{p_k} (= C_k)$ [J/K]	$h_k A_k (= H_k)$ [W/K]	Δq_k [W]	RMSE [°C]
CPU	78.40	2.29	67.54	1.13
GPU	71.83	0.87	39.48	0.53

V. RESULTS AND DISCUSSIONS

A. Parameter estimation

First, the validity of the proposed model is verified through parameter estimation on a real machine. To obtain the estimate, the analytical solutions of equations (4)–(7) are derived and fitted with the measured temperature curve.

We use equation (7) to estimate the parameters related to the devices in the server. To simplify the analytical solution, the stress-ng command and GPU burn are used to provide a particular load. When a particular load is applied at $t > 0$, the power consumption of the hardware q_k , and the temperature T_k can be obtained from equation (7).

The least squares method is used to fit the curve, as shown in Figure 7, which shows the comparison of the predicted temperature curve and the measured values of the CPU and GPU. Table 2 lists the results of the parameter estimation and root mean square error (RMSE) of the predicted and estimated values. The values in the table represent the average of 17 experiments.

As can be seen from Figure 5, the GPU temperature increase can be described with a high accuracy. At the same time, the RMSE value of the CPU is relatively high, and a deviation from the actual temperature curve is observed. Figure 8(a) shows a comparison between the predicted temperature profile in this case and the measured value at $t < 200$. As can be seen from the figure, compared with the estimated value of the model, the CPU value has a gradual slope from the middle of the temperature rise. It is because the CPU parameters vary due to factors such as the CPU's local fan and clock frequency control, which makes heat emission easier and is not precisely modeled in the proposed model. To simplify the model, the parameters are divided into low-temperature and high-temperature values, and a constant temperature is assumed as the threshold value.

When determining the threshold for dividing the parameters, a random matrix calculation program is used to provide a low load that maintains the maximum temperature of the CPU at 45 °C, and the curve fitting estimates the heat transfer coefficient at low temperatures. Next, to estimate the range of the heat transfer coefficient at a low temperature, when the CPU load is 100%, t as a parameter change boundary is fitted from 1 s to 200 s. According to the fitting results, we estimate t , which minimizes the RMSE between the estimated and measured values and determines $h_c A_{cpu}$ at high temperatures. Figure 8(b) shows a comparison of the estimated temperature curve and the measured values for $t < 200$ after the parameter division. Compared with the previous values of the segmentation parameters (Table 2), the

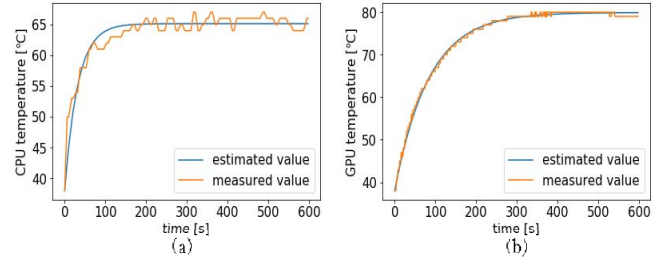
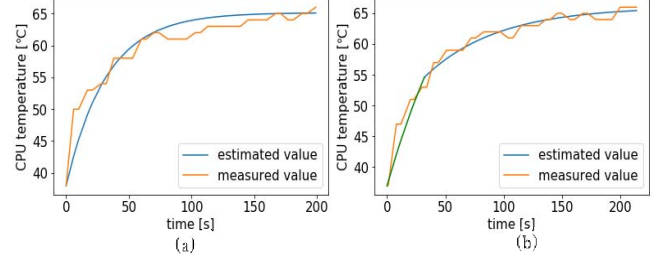


Fig.7. (a) CPU (b) GPU



(a) Estimation result with a single parameter
(b) Estimation result with a parameter division

Fig.8. Comparison of estimated temperature curve and measured values at $t < 200$

TABLE 3 COMPARISON OF THE EXISTING METHOD AND THE PROPOSED METHOD

application	CPU1	CPU2	GPU
stress-ng CPU100%	82.4	-	-
stress-ng CPU40%	30.3	-	-
matrix operations	-	30.2	44.5
gpu-burn	-	36.1	62.8
gpu-burn +stress-ng CPU100%	82.4	36.1	62.8

predicted value of the RMSE remains low, and it can be seen that more suitable parameters are obtained.

Next, equations (4) to (6) are used to estimate the parameters of the server enclosure and air intake.

B. Server exhaust temperature prediction

For evaluating the accuracy of the proposed temperature prediction model, the exhaust temperatures for different loads were predicted. To confirm the prediction accuracy when applying loads with different resource utilization rates, five different types of power loads were applied to the server model with two CPUs and one GPU. The load in each case was 10 min, and the processing content was constant during this period. In addition, the temperature data between the 10 min of the given load and 5 min after the idle state are used to evaluate the prediction accuracy. Table 3 lists five applications for the temperature prediction and power load for each resource. In addition to the above stress-ng and GPU burn, we used CUDA to perform random matrix operations to provide a low load for the CPU and GPU.

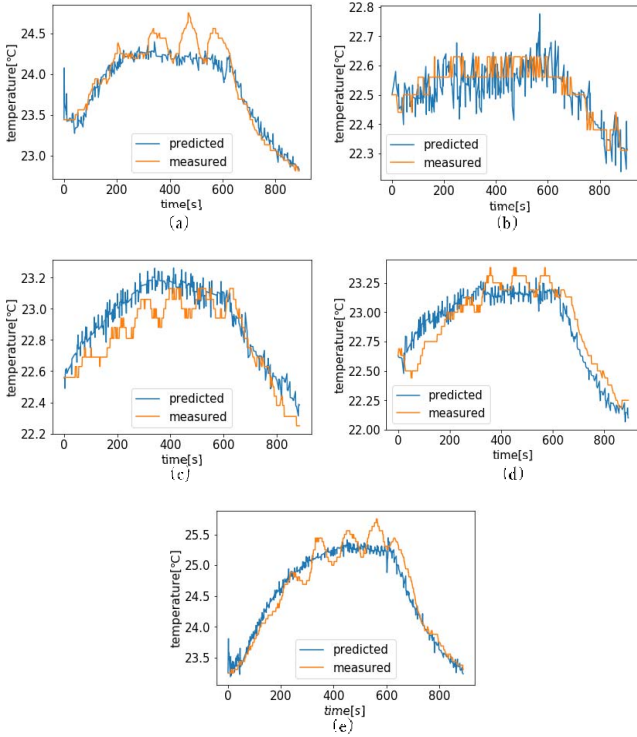


Fig.9. Comparison of predicted and measured values of server exhaust temperature

In addition, in the parameter estimation experiment, for the sake of simplicity, the server intake air temperature and power consumption are regarded as constants, and the parameters are determined by taking the average value of many experiments. However, in practice, it is assumed that the resource consumption is a variable. Therefore, equations (4) to (7) are discretized, and the real-time measured server inlet air temperature and CPCP/GPU power are used for sequential predictions. When there are three heat sources, the discrete prediction model is as follows: Starting from the actual hardware measurement position, λ_1 , λ_2 , and λ_2 are set as 0, 0.33, and 0.17, respectively.

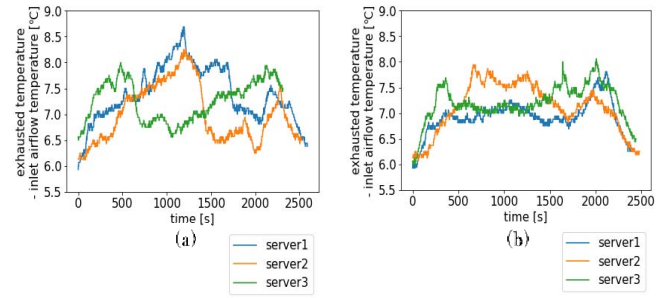
TABLE 4 COMPARISON OF THE EXISTING METHOD AND THE PROPOSED METHOD

load	RMSE (Average of 5 times) [°C]
stress-ng CPU100%	0.23
stress-ng CPU40%	0.11
matrix operations	0.22
gpu-burn	0.20
gpu-burn +stress-ng CPU100%	0.22

Figure 9 shows a comparison of the predicted and measured values for the five loads. Table 4 shows the RMSE values for each load prediction performed five times. It can be seen from the above that even if different loads are applied to multiple heat sources, the proposed method can predict the exhaust temperature of the server with high accuracy.

C. Load balancing simulation

Then, the load-balancing method is evaluated through a simulation. At this time, the requests that were originally



(a) Minimum CPU load (b) Proposed method
Fig.10. Temperature rise of distribution destination server

TABLE 5 PREDICTION ACCURACY OF SERVER EXHAUST TEMPERATURE IN THE PROPOSED METHOD

Method	Mean value of maximum temperature rise [°C]	Mean value of temperature variance [°C^2]
Minimum CPU load	7.71	0.287
Proposed method	7.44	0.096

executed by multiple servers in parallel are executed in series in the one server used in the model construction; thus, the simulation is carried out. In the experiment, the server inlet temperature and resource power consumptions in the execution of each request are unchanged. The prediction of the discharge temperature and the determination of the target server are conducted before the request is executed. To verify the effect of the temperature prediction, the proposed method is compared with the existing minimum CPU load, which allocates the load after comparing the CPU utilization of each server and selecting the server with the minimum utilization.

In addition, the simulation load is generated based on Google cluster workload traces [25]. The content of the request is randomly selected from the previous five types of loads. Three servers with previously derived parameters are used for simulation and comparison.

Figure 10 shows the results when the simulation is performed at the same request arrival frequency as the original data. The figure shows the difference between the exhaust and inlet temperatures of the three load balancing servers using the existing method and the proposed method. As can be seen from the figure, the temperature increase between servers varies less in the proposed approach than with the minimum CPU load. Table 5 compares the variance of the maximum exhaust and exhaust air temperatures of the three servers in the existing and proposed methods. The proposed method can reduce the maximum temperature rise to 0.27 °C lower than the existing methods, and it can be seen that the temperature variance at the same time of the three servers is reduced by 67% on average.

From the above results, we consider the effectiveness of the proposed method from the perspective of cooling efficiency. It can be clearly seen from equation (4) that the difference between the exhaust temperature and the intake air temperature is constant in steady state. Therefore, by continuously reducing the maximum exhaust temperature of the server group by 1 °C, the set temperature of CRAC can be increased by 1 °C. According to the existing research, the relationship between the set temperature and COP value of a typical water-cooled CRAC is as follows.

$$y_{cop} = 0.0068x^2 + 0.0008x + 0.4580, \quad (8)$$

where y_{cop} is the COP value of the CRAC, and x is the set temperature. According to this equation, compared with the existing methods [9], the CRAC COP value for the proposed method is increased by 0.063. As the COP value is the cooling effect of the CRAC per 1 kW, the cooling effect of the CRAC is increased by 2.5%, and the cooling power is reduced by 2.6%. According to a report by Emerson Network Power, a US data center operator, a typical 5000 square foot data center consumes 10,296 kWh per day for cooling [26]. Therefore, by using the proposed method, the cooling power consumption of the CRAC is reduced by 2.6%, and the cooling power consumption of the data center is reduced by 260 kW per day, that is, 95 MW per year.

VI. CONCLUSION

In this study, we constructed a simple and general theoretical model to describe the temperature change in a server with multiple heat sources and proposed a load balancing method considering the predicted value of the server exhaust temperature through the model. In this study, we evaluated the accuracy of the proposed temperature model on a real server and evaluated the proposed load balancing method through simulations. According to the results, in the proposed temperature model, when different loads are applied to two CPUs, and GPUs are assumed to be heat sources in the server, the RMSE is in the range of 0.11 to 0.23, and the exhaust temperature of the server can be predicted effectively. In the proposed load balancing method, we compared the proposed method with the minimum CPU load method and reduced the exhaust temperature variance between servers. By using the proposed method and reducing the maximum temperature of the server by 0.27 °C, the academic data center can reduce the cooling power by 260 kW per day and 95 MW per year.

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